

Lesson Objective

Plot data on line graphs and analyze trends.

1. Plot data from a table on a line graph to represent change over time.

The table shows how much Cami weighed at each 4-month period. Plot each point on the coordinate plane, then connect the points with line segments to make a line graph.

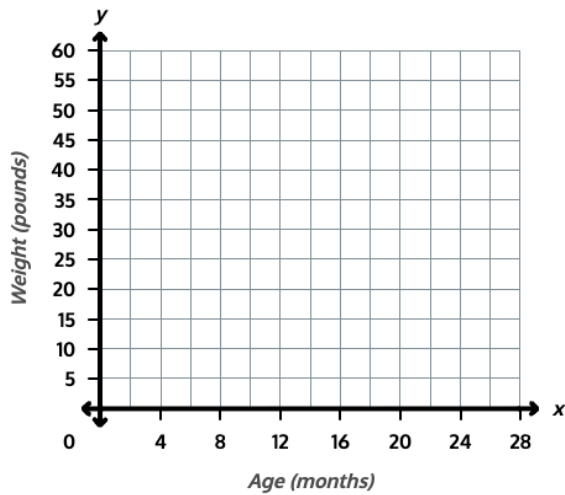
Age (months)	Weight (pounds)
0	5
4	20
8	30
12	40
16	45
20	50
24	50
28	50

Show students the table and coordinate plane. Ask students what the x-axis and y-axis show, and how the axes are labeled.

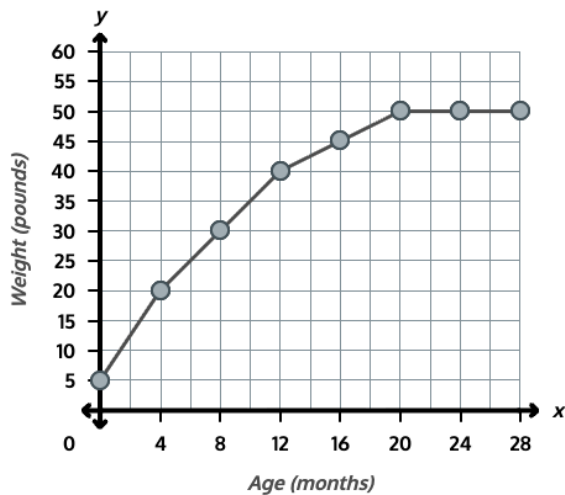
Model plotting the first point. Tell students that at 0 months, Cami weighed 5 pounds, so the point is at (0, 5). Then model plotting (4, 20) by counting over 4 months and up to 20 pounds.

Have students plot the remaining points from the table on their own coordinate planes.

Guide students to connect each pair of points with a line segment to form the line graph. Ask students what the line shows about how Cami's weight changed over time.



Answer:



2. Analyze the line graph to find differences between data points and compare rates of change.

Use Cami's weight graph from Problem 1 to answer each question.

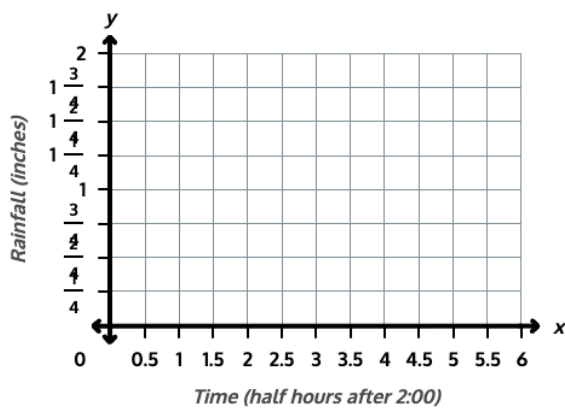
Ask students to find Cami's weight at 0 months and at 4 months on the graph. Have students subtract $20 - 5$ to find she gained 15 pounds.

Have students find the difference between her weight at 4 months and at 8 months. Guide them to subtract $30 - 20 = 10$ pounds.

a. How much weight did Cami gain in the first 4 months of her life? b. How much weight did Cami gain between 4 months and 8 months? c. Did Cami gain more weight in the first 4 months or the second 4 months? How does the line graph show this? <i>Answer: a) 15 pounds; b) 10 pounds; c) The first 4 months; the line segment from 0 to 4 months is steeper than the segment from 4 to 8 months.</i>	Ask students to compare 15 pounds and 10 pounds. Have students look at the two line segments and describe what they notice about their steepness. Point out that the segment from 0 to 4 months is steeper because Cami gained more weight in the same amount of time, so the line rises more sharply.
3. Plot a new data set on a line graph and interpret trends, including intervals of no change. The table shows the total rainfall measured each half hour during a storm. Plot each point on the coordinate plane and connect them with line segments. Then answer the questions.	Have students plot each point from the table and connect the points with line segments. Check that students count carefully along the x-axis (each tick is one half hour) and along the y-axis (each tick is $\frac{1}{4}$ inch). Ask students to look across the graph and find the steepest segment. Have them subtract $1\frac{3}{4} - 1$ to find that $\frac{3}{4}$ inch of rain fell between 4:30 and 5:00. Point out that the steeper the segment, the more rain fell during that interval. Ask students what they notice about the line between 3:30 and 4:30. Guide students to see that the line is flat, or horizontal, because the total rainfall stayed at 1 inch. Tell students that a

Time Total Rainfall (inches)

2:00	0
2:30	$\frac{1}{4}$
3:00	$\frac{3}{4}$
3:30	1
4:00	1
4:30	1
5:00	$1\frac{3}{4}$



a. During which half hour did the most rain fall?

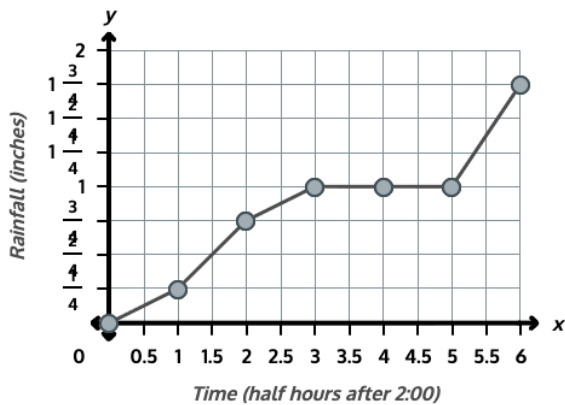
How can you tell from the graph?

b. What happened to the total rainfall between 3:30 and 4:30? How does the graph show this?

Answer: a) Between 4:30 and 5:00; this segment is the steepest, and the rainfall increased by $\frac{3}{4}$ inch ($1\frac{3}{4} - 1 = \frac{3}{4}$); b) The total rainfall stayed at

horizontal segment means no change happened during that interval, so no rain fell.

1 inch, so no rain fell; the line is horizontal between those times.



Monitor For

- Do students read the scaled axes correctly, counting by 4 months on the x-axis and by 5 pounds on the y-axis in Problem 1?
- Do students subtract the y-values at two points to find the difference in weight, rather than counting the number of grid lines?
- Do students connect the steepness of a segment to the rate of change in Problems 2 and 3, or do they only describe the line as "going up"?

Instructional Moves

- Before plotting in Problem 1, have students trace one finger along the x-axis and one along the y-axis to confirm each labeled interval (every 4 months, every 10 pounds).
- If a student counts grid lines instead of subtracting y-values, prompt them to name the weight at each point first, then write a subtraction equation such as $30 - 20 = 10$.
- Emphasize the link between steepness and rate of change by asking, "Which segment rises more in the same amount of time?" when comparing intervals in Problems 2 and 3.

- Do students recognize that a horizontal segment means the quantity did not change during that interval?

- If a student says the rain stopped because the line went down, point out that the line is flat, not going down, and ask what the y-value is at each end of that segment.

Reflection Questions

- How did you use the table to plot each point on the line graph in Problem 1? *Answer: I matched the age in the table to the x-coordinate and the weight to the y-coordinate, then plotted the point where they met.*

- How did you find how much weight Cami gained between two ages? *Answer: I subtracted her earlier weight from her later weight. For example, $30 - 20 = 10$ pounds between 4 and 8 months.*

- What does a steeper line segment tell you about the change between two points? *Answer: A steeper segment means the quantity changed more during that interval, so the rate of change was faster.*

- What does a horizontal line segment on a line graph show? *Answer: It shows that the quantity did not change during that interval, like when no rain fell between 3:30 and 4:30.*